8.2.37

AI25BTECH11003 - Bhavesh Gaikwad

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Question

Find the equation of the conic, that satisfies the given conditions. vertex (-3, 0), directrix $x + 5 = 0$.

Theoretical Solution

Given conditions:
Vertex: $(-3, 0)$
Directrix: $x + 5 = 0$

$\Rightarrow$ Since we have a vertex and directrix, this conic is a parabola with eccentricity $e = 1$.

Using the formulation for a conic with directrix $\mathbf{n}^T\mathbf{x} = c$, eccentricity e, and focus $\mathbf{F}$:

Given Parameters:
Normal vector: $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
Directrix constant: $c = -5$
Eccentricity: $e = 1$

Theoretical Solution

Focus: $\mathbf{F} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ [derived from equidistance property]

Calculating matrix $\mathbf{V}$

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2(\mathbf{n}\mathbf{n}^T) = \mathbf{I} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{1}$$

Calculating $\mathbf{u}$

$$\mathbf{u} = ce^2\mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F} = (-5)(1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} - 1 \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} \tag{2}$$

Calculating f:

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2e^2 = 1 - 25 = -24 \tag{3}$$

Theoretical Solution

$$\text{The eigenvalues of } \mathbf{V} \text{ is } \lambda_1 = 0 \tag{4}$$

Since the eigenvalue is zero, we confirm this is a parabola ($e = 1$).
The general conic equation is:

$$\mathbf{x}^T\mathbf{V}\mathbf{x} + 2\mathbf{u}^T\mathbf{x} + f = 0 \tag{5}$$

Theoretical Solution

Substituting our values in Equation 5:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} -4 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + (-24) = 0 \tag{6}$$

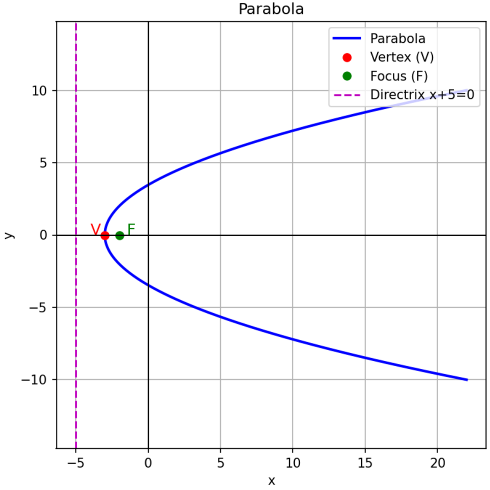
We Get,

$$\boxed{y^2 - 8x - 24 = 0} \tag{7}$$

Therefore, the Equation of the Conic is:

$$\boxed{y^2 = 8(x + 3)} \tag{8}$$

Parabola



Parabola
Parabola
Vertex (V)
Focus (F)
Directrix x+5=0
V
F
10
5
0
−5
−10
y
−5
0
5
10
15
20
x